

Mary Maceda^{1*}, Victor Duraes de Faria¹, Chris Vermillion², Anderson Rodrigo de Queiroz¹
¹ North Carolina State University, ² University of Michigan, * Correspondence: memaceda@ncsu.edu

Background

- ❖ **NC House Bill 951:** 70% reduction in CO2 emissions by 2030 and carbon neutrality by 2050 (GANC, 2021).
- ❖ **NC EO 218:** Offshore wind development goals of 2.8 GW by 2030 and 8.0 GW by 2040 (SNC, 2021).
- ❖ NC has ample offshore wind (NREL, 2016) and ocean current potential (GTRC, 2013). **Can we coordinate the generation from these resources to achieve economic benefits?**

Aim

- ❖ Develop models to coordinate offshore wind and ocean current resources, improving energy density and levelized cost of energy [\$/MWh], identifying promising device designs and site locations.

Project Diagram

Data

- **Wind Speed** Data at 100, 140, 160m Height; From 2007-2013 1h Time Discretization; 2x2km (NREL, 2015)
- **Ocean Current** Data For Entire Water Column; From 2007-2013; 3h Time Discretization; 1/12° Resolution (HYCOM, 2023)
- NREL ATB23 Data For **Wind Turbine Costs** – Depth Dependent (NREL, 2023)
- **Kite Component Costs** (ARPA-E, 2020)
- **Transmission System Component Costs:** Cables, Substations, Reactive Power Compensation (Faria, 2023)

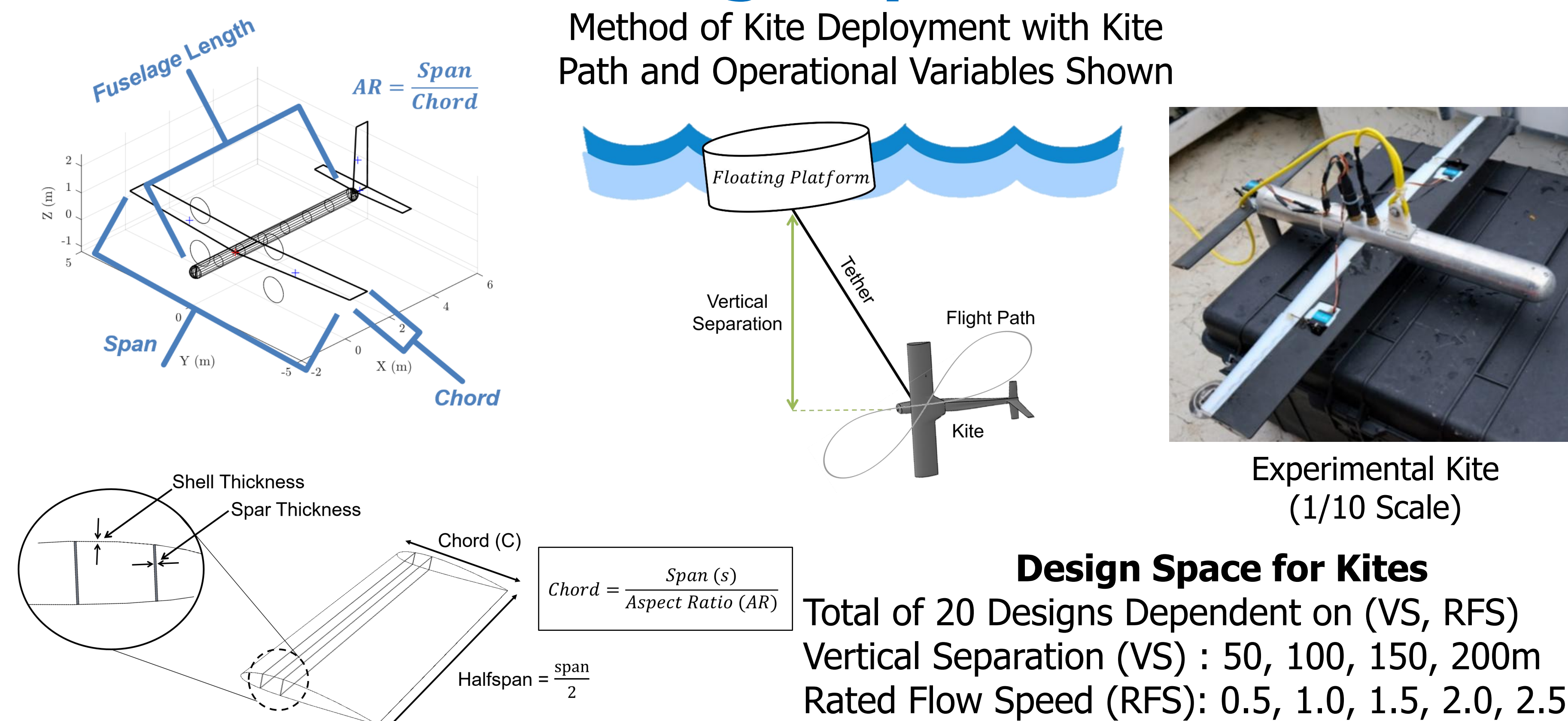
Models/Methods

- **Kite Design Optimization Model**
- **Transmission System Optimization Model** (HVDC/HVAC)
- Modules for Computing **Power and Annualized Costs** at Individual Site Locations for Kites and Wind Devices
- Modules for **Changing Time and Spatial Resolution** to Improve Computational Performance
- **Portfolio Optimization Model:** Define Optimal Locations, Device Type, Deployment Size, and Transmission System Location

- **Investigate Different Integrations of Wind and Kite Devices, Considering Improvements in the Utilization of Existing System Infrastructure**

Kite Design Optimization

Method of Kite Deployment with Kite Path and Operational Variables Shown

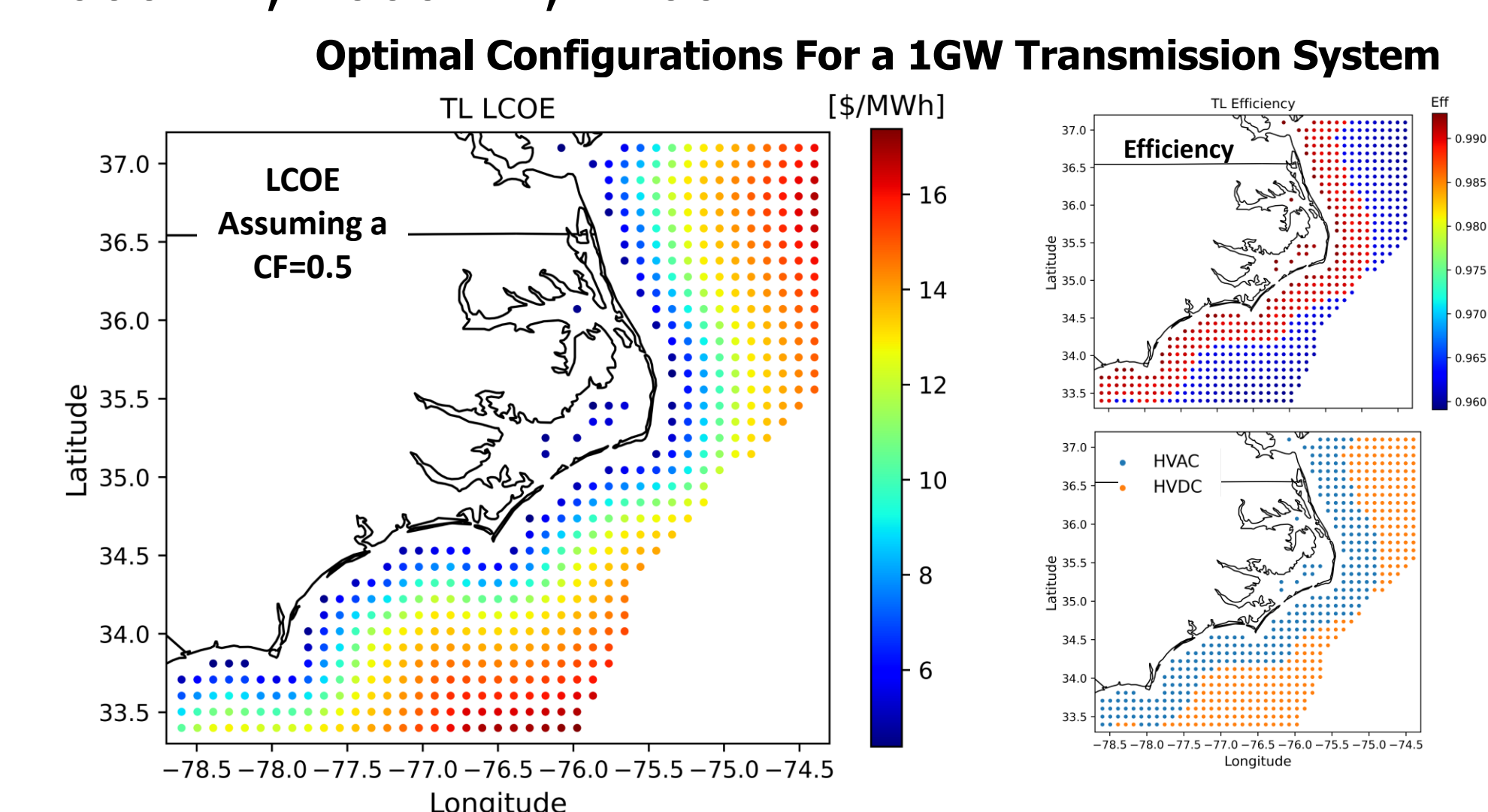


Portfolio Optimization Model (Simplified)

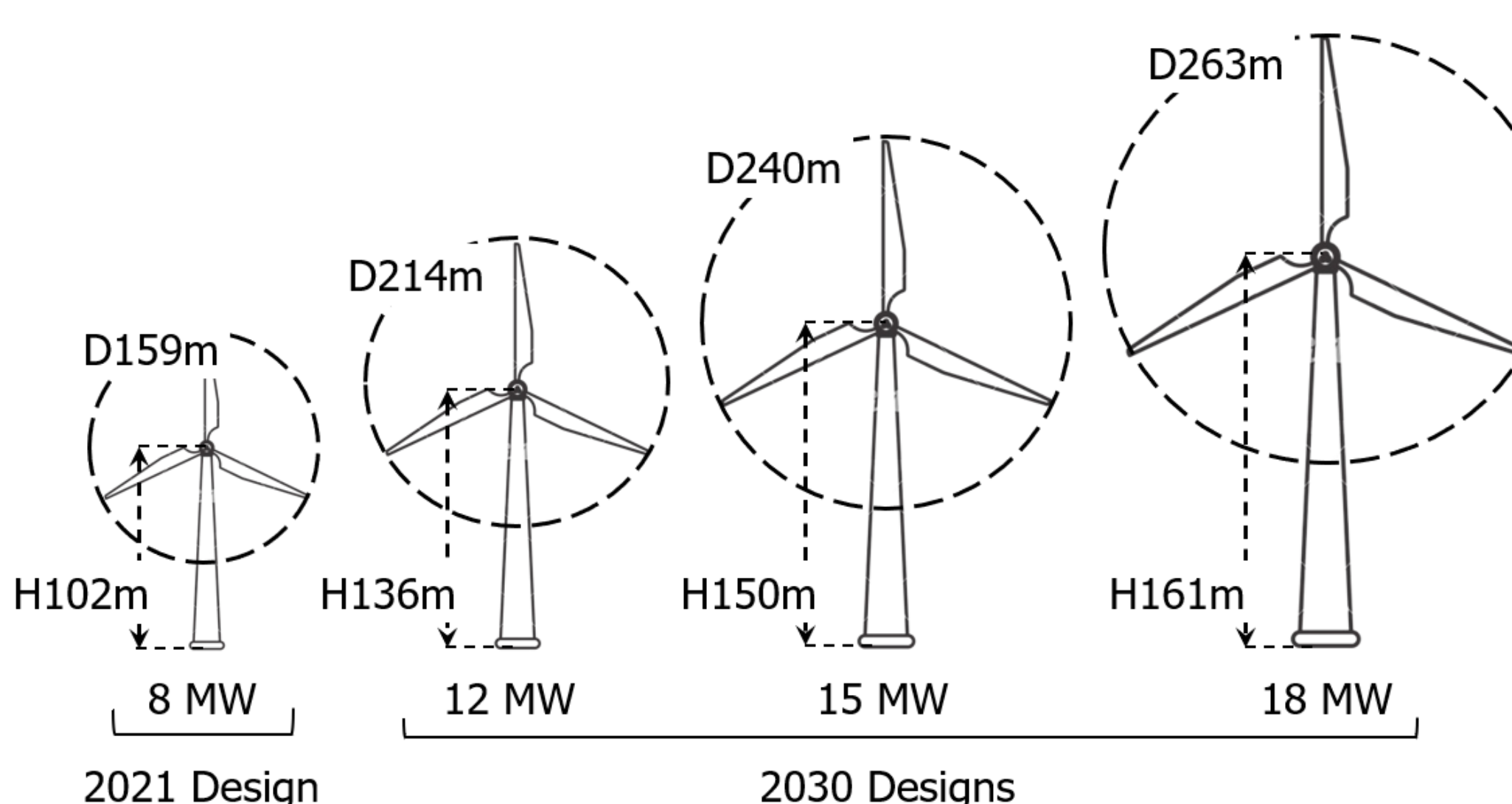
$$\begin{aligned} & \text{Max}_{(y,w,v)} \sum_{t \in T} (EG_t^{Wind} + EG_t^{Kite} - \Delta_t) & (1) \\ \text{s.t.} & \frac{TAC^{Wind} + TAC^{Kite} + TC}{8760 \sum_{t \in T} (EG_t^{Wind} + EG_t^{Kite} - \Delta_t)} \leq \overline{LCOE} & (2) \\ & TAC^{Wind} = \sum_{i \in I_{Wind}} AC_i^{Wind} y_i^{Wind} & (3) \\ & TAC^{Kite} = \sum_{i \in I_{Kite}} \sum_{d \in D} AC_{i,d}^{Kite} y_{i,d}^{Kite} & (4) \\ & TC = \sum_{i \in I_{Tr}} v_i TC_i & (5) \\ & EG_t^{Wind} = \sum_{i \in I_{Wind}} EG_{i,t}^{Wind} y_i^{Wind} & \forall t \in T & (6) \\ & EG_t^{Kite} = \sum_{i \in I_{Kite}} \sum_{d \in D} EG_{i,d,t}^{Kite} y_{i,d}^{Kite} & \forall t \in T & (7) \\ & \sum_{i \in I_{Kite}} y_{i,d}^{Kite} \leq w_d \overline{NT}^{Kite} & \forall d \in D & (8) \\ & \sum_{d \in D} w_d^{Kite} \leq \overline{ND}^{Kite} & (9) \\ & EG_t^{Wind} + EG_t^{Kite} - \Delta_t \leq RP_{Tr} & \forall t \in T & (10) \\ & y_i^{Wind} \leq \overline{NT}_i^{Wind} & \forall i \in I_{Wind} & (11) \\ & \sum_{d \in D} y_{i,d}^{Kite} \leq \overline{NT}_i^{Kite} & \forall i \in I_{Kite} & (12) \\ & \sum_{k \in OR_{v_i}^{Wind}} y_i^{Wind} + \sum_{(k,d) \in OR_{v_i}^{Kite}} y_{i,d}^{Kite} \leq (1 - v_i) \overline{NT} & \forall i \in I_{Tr} & (13) \\ & \sum_{i \in I_{Tr}} v_i = 1 & (14) \end{aligned}$$

Transmission System Optimization

Design Space for Transmission System
 Maximum Active Power Capacity: 100MW, 300MW, 600MW, 1000MW, 1200MW

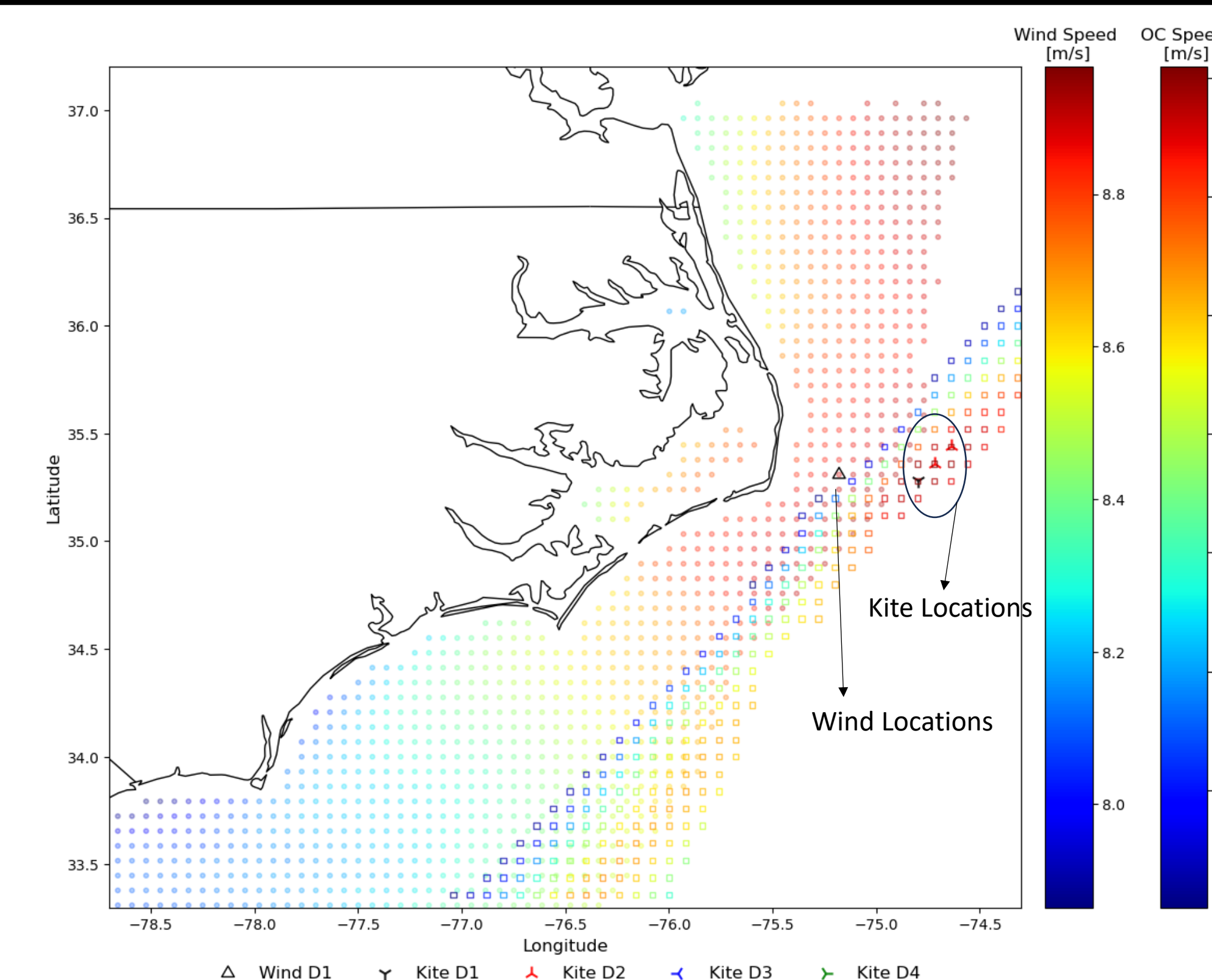
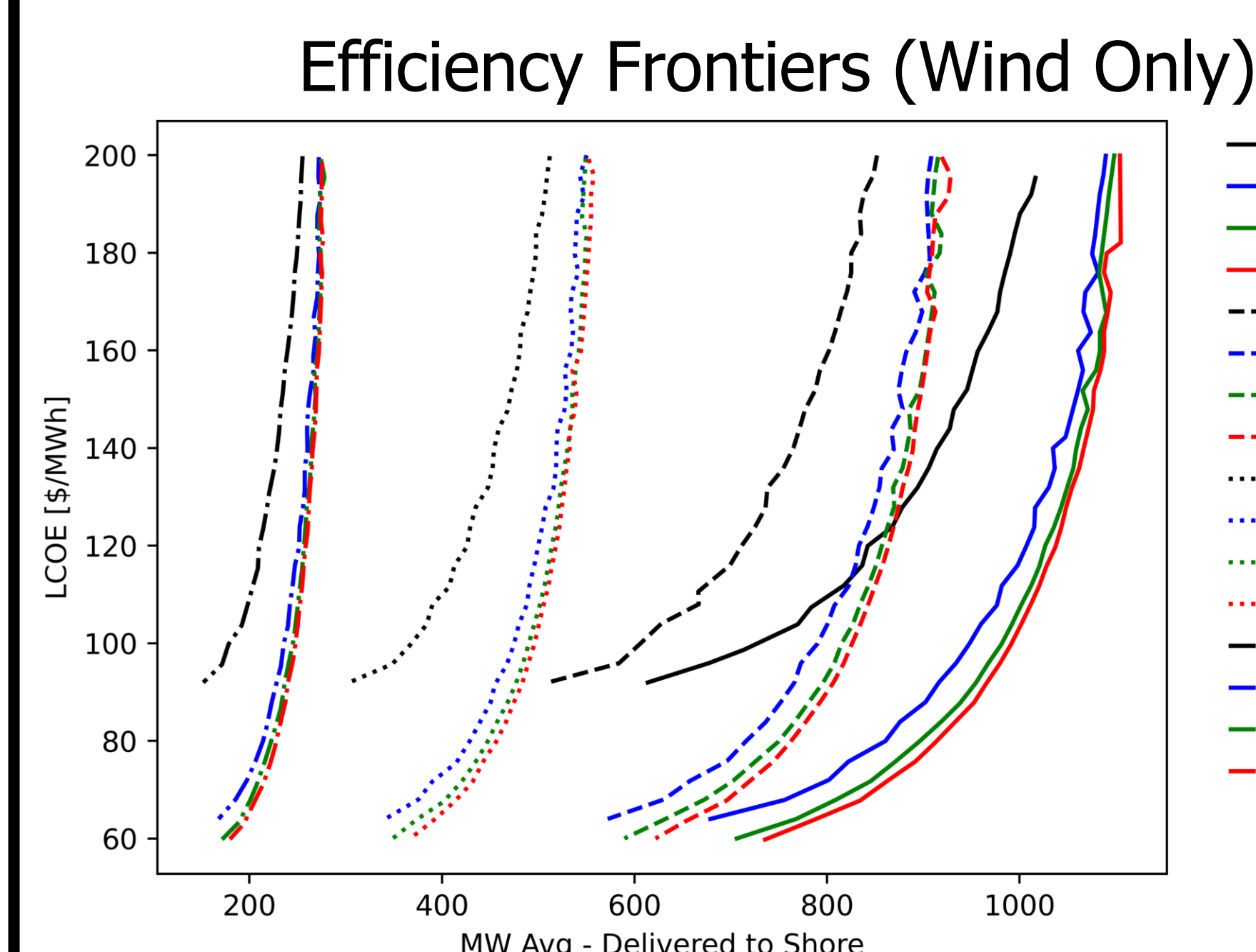


Wind Turbines Investigated: ATB-2023 (NREL, 2023)



Model Explanation: For a limited budget (2) ($\overline{LCOE}$), we want to find the optimal number of turbines per site location ($y_i^{Wind}, y_{i,d}^{Kite}$) such that the amount of energy we deliver to shore is maximized (1), in this process the model must account for energy curtailment (10), maximum packing density of each site (11, 12), limits in the radius of the energy collection system (13,14), and system component costs (3, 4, 5).

Preliminary Results



Coordinating Wind and Kite Designs for a Min LCOE of 40 [\$/MWh] on a 1GW TL. Wind (D1-18MW), Kite Design For 50m of Vertical Separation (D1-2.5m/s, D2-2.0m/s, D3-1.5m/s, D4- 1.0m/s)

Best Solution Wind Alone:
 LCOE : 60[\$/MWh],
 Energy: 720[MW Avg]

Best Solution Wind-Kites:
 LCOE : 40[\$/MWh]
 Energy: 620[MW Avg] (2/3 Kites)

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